Chapter 5. The Riemann–Stieltjes Integral

Riemann integral, integrability criteria, fundamental theorem, bounded variation, Stieltjes integrals

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5.1 The Riemann integral

A **partition** of $[a,b]$ is a finite set $P = \{a = x_0 < x_1 < \cdots < x_n = b\}$; write $\Delta x_i = x_i - x_{i-1}$ and $\|P\| = \max \Delta x_i$ (the mesh). $Q \supset P$ is a **refinement** of P. For bounded $f : [a,b] \to \mathbb{R}$ let $M_i = \sup_{[x_{i-1},x_i]} f$, $m_i = \inf_{[x_{i-1},x_i]} f$, and

$$U(f,P) = \sum M_i \Delta x_i, \qquad L(f,P) = \sum m_i \Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int} f = \inf_P U(f,P)$ and $\underline{\int} f = \sup_P L(f,P)$; f is **Riemann integrable** ($f \in \mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P \subset Q$, then $L(f,P) \le L(f,Q) \le U(f,Q) \le U(f,P)$. For any partitions P, P', $L(f,P) \le U(f,P')$; hence $\underline{\int} f \le \overline{\int} f$.

Proof. It suffices to add one point $y \in (x_{i-1}, x_i)$: the infima of f over $[x_{i-1}, y]$ and $[y, x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P) \le L(f, P \cup P') \le U(f, P \cup P') \le U(f,P')$. ■

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon > 0$ there is a partition P with $U(f,P) - L(f,P) < \epsilon$.

Proof. If the condition holds, $0 \le \overline{\int} f - \underline{\int} f \le U(f,P) - L(f,P) < \epsilon$. Conversely, if $f \in \mathcal{R}$, choose P_1, P_2 with $U(f,P_1) < \int f + \epsilon/2$ and $L(f,P_2) > \int f - \epsilon/2$; then $P = P_1 \cup P_2$ works by Proposition 5.1. ■

Theorem 5.3 (5.1.3)

If $f, g \in \mathcal{R}$ and $c \in \mathbb{R}$, then $f+g, cf \in \mathcal{R}$ with $\int (f+g) = \int f + \int g$ and $\int cf = c \int f$; and $f \le g$ implies $\int f \le \int g$.

Proof. On each subinterval $\sup(f+g) \le \sup f + \sup g$ and $\inf(f+g) \ge \inf f + \inf g$, so

$$L(f,P) + L(g,P) \le L(f+g,P) \le U(f+g,P) \le U(f,P) + U(g,P).$$

Choosing P with $U - L < \epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c \ge 0$, $U(cf,P) = cU(f,P)$ and $L(cf,P) = cL(f,P)$; for $c = -1$, $U(-f,P) = -L(f,P)$. If $f \le g$, then $U(f,P) \le U(g,P)$ for all P. ■

Products of integrable functions are integrable: $f^2 \in \mathcal{R}$ because on each subinterval $\sup f^2 - \inf f^2 \le 2K(M_i - m_i)$ with $K = \sup|f|$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f \in \mathcal{R}[a,b]$ and $a < c < b$, then $f \in \mathcal{R}[a,c]$, $f \in \mathcal{R}[c,b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. Conversely, integrability on $[a,c]$ and $[c,b]$ implies integrability on $[a,b]$.

Proof. For partitions P of $[a,b]$ containing c, with $P_1 = P \cap [a,c]$ and $P_2 = P \cap [c,b]$, we have $U(f,P) = U(f,P_1) + U(f,P_2)$, likewise for L. If $U(f,P) - L(f,P) < \epsilon$ (we may add c to P), the same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $\left|\int_a^b f\right| \le \int_a^b |f|$.

Proof. From $||f(x)| - |f(y)|| \le |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|, P) - L(|f|, P) \le U(f, P) - L(f, P)$. Then $-|f| \le f \le |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0, 1]$: every $U(f, P) = 1$ and every $L(f, P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Theorem 5.6 (5.2.1)

Every continuous $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b - a)$ when $|x - y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b-a)$ (the extrema are attained), so $U(f, P) - L(f, P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a, b]$ into n equal parts. Then $M_i = f(x_i)$, $m_i = f(x_{i-1})$, and $U(f, P) - L(f, P) = \frac{b-a}{n} \sum (f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) < \epsilon$ for large n. ■

More generally, a bounded function with finitely many discontinuities is integrable (enclose them in short intervals), and so is one whose discontinuity set has measure zero (Chapter 10 of the second volume).

A **Riemann sum** is $S(f, P) = \sum f(t_i)\Delta x_i$ with $t_i \in [x_{i-1}, x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A \in \mathbb{R}$ the following are equivalent:
(a) $f \in \mathcal{R}$ and $\int_a^b f = A$;
(b) for every $\epsilon > 0$ there is $\delta > 0$ with $|S(f, P) - A| < \epsilon$ for every partition with $\|P\| < \delta$ and every choice of the t_i.

Proof. (a)⇒(b): let $K = \sup|f|$ and choose P_0, with k points, such that $U(f, P_0) - L(f, P_0) < \epsilon/2$. Let $\|P\| < \delta = \epsilon/(8Kk)$. Passing from P to $P \cup P_0$ changes only the at most k subintervals of P that contain points of P_0 in their interior, and each change of U or L is at most $2K\|P\|$. So $U(f, P) - L(f, P) \le U(f, P \cup P_0) - L(f, P \cup P_0) + 4Kk\delta < \epsilon$. Since both $S(f, P)$ and A lie in $[L(f, P), U(f, P)]$, $|S(f, P) - A| < \epsilon$.

(b)⇒(a): fix P with $\|P\| < \delta$. Choosing the t_i with $f(t_i) > M_i - \epsilon/(b - a)$ gives $U(f, P) < S(f, P) + \epsilon < A + 2\epsilon$; similarly $L(f, P) > A - 2\epsilon$. So $U(f, P) - L(f, P) < 4\epsilon$, and $\int f = A$. ■

Example 5.2

$\int_0^1 x^2\,dx = \lim_n \frac{1}{n}\sum_{k=1}^n (k/n)^2 = \lim \frac{(n+1)(2n+1)}{6n^2} = \frac{1}{3}$. Also $\lim_n \sum_{k=1}^n \frac{1}{n+k} = \int_0^1 \frac{dx}{1+x} = \log 2$.

5.3 The fundamental theorem of calculus

For $f \in \mathcal{R}[a,b]$ let $F(x) = \int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0) = f(x_0)$.

Proof. With $K = \sup|f|$, $|F(y)-F(x)| = |\int_x^y f| \le K|y-x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t)-f(x_0)| < \epsilon$ for $|t-x_0| < \delta$; then for $0 < |h| < \delta$,

$$\left|\frac{F(x_0+h)-F(x_0)}{h} - f(x_0)\right| = \left|\frac{1}{h}\int_{x_0}^{x_0+h} (f(t)-f(x_0))\,dt\right| \le \epsilon.$$

■

Corollary 5.10 (5.3.2)

Every continuous $f : [a,b] \to \mathbb{R}$ has a primitive, namely $F(x) = \int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F : [a,b] \to \mathbb{R}$ is differentiable and $f = F'$ is integrable, then $\int_a^b f = F(b) - F(a)$.

Proof. For any partition, the mean value theorem gives $t_i \in (x_{i-1}, x_i)$ with $F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i$, so $F(b) - F(a) = \sum f(t_i)\Delta x_i \in [L(f,P), U(f,P)]$. As $f \in \mathcal{R}$, $F(b) - F(a) = \int f$. ■

The integrability hypothesis cannot be dropped: there are differentiable functions with bounded, non-integrable derivatives (Volterra), and $x^2\sin(1/x^2)$ has an unbounded derivative on $[0,1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g : [a,b] \to [c,d]$ is a surjective C^1 function and $f : [c,d] \to \mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(t))g'(t)\,dt$.

Proof. Let F be a primitive of f (Corollary 5.10). By the chain rule $(F \circ g)' = (f \circ g)g'$, which is continuous, hence integrable, so by Theorem 5.11 the right side is $F(g(b)) - F(g(a))$, which is also the left side. ■

Proposition 5.13 (5.3.5, integration by parts)

If $F, G : [a,b] \to \mathbb{R}$ are differentiable and $f = F'$, $g = G'$ are integrable, then $\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG$.

Proof. F, G are continuous, hence integrable, so $(FG)' = fG + Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ■

5.4 Functions of bounded variation

For $f:[a,b]\to\mathbb{R}$ and a partition P, let $V(f,P)=\sum|f(x_i)-f(x_{i-1})|$. The **total variation** is $V_a^b(f)=\sup_P V(f,P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f,P)$ (triangle inequality). Monotone functions have $V_a^b(f)=|f(b)-f(a)|$; Lipschitz functions with constant K have $V_a^b(f)\le K(b-a)$; BV functions are bounded, $|f(x)|\le|f(a)|+V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f,g are of bounded variation, then so are $f+g$ and fg, with

$$V(f+g)\le V(f)+V(g),\qquad V(fg)\le\|f\|_\infty V(g)+\|g\|_\infty V(f).$$

If moreover $|f|\ge c>0$, then $1/f$ is of bounded variation with $V(1/f)\le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i)-(f+g)(x_{i-1})|\le|\Delta f_i|+|\Delta g_i|$, and

$$|f(x_i)g(x_i)-f(x_{i-1})g(x_{i-1})|\le|f(x_i)||\Delta g_i|+|g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i)-1/f(x_{i-1})|=|\Delta f_i|/|f(x_i)f(x_{i-1})|\le|\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a<c<b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f)=V_a^c(f)+V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P\ni c$ of $[a,b]$, $V(f,P)=V(f,P\cap[a,c])+V(f,P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f=g-h$ with g,h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x)=V_a^x(f)$ (with $g(a)=0$) and $h=g-f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\ge 0$. ∎

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha)=\sup_P\sum\|\alpha(x_i)-\alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha=(\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha)<\infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i)-\alpha_k(x_{i-1})|\le\|\alpha(x_i)-\alpha(x_{i-1})\|\le\sum_j|\alpha_j(x_i)-\alpha_j(x_{i-1})|$. Summing, $V(\alpha_k,P)\le\sum_i\|\Delta\alpha_i\|\le\sum_j V(\alpha_j,P)$; take suprema. ∎

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha)=\int_a^b\|\alpha'(t)\|\,dt$.

Proof. For vector-valued v (integrated componentwise), $\|\int v\|\le\int\|v\|$: with $u=\int v$, $\|u\|^2=\int\langle u,v\rangle\le\|u\|\int\|v\|$ by Cauchy–Schwarz. Hence $\|\alpha(x_i)-\alpha(x_{i-1})\|=\|\int_{x_{i-1}}^{x_i}\alpha'\|\le\int_{x_{i-1}}^{x_i}\|\alpha'\|$, and $\Lambda(\alpha)\le\int\|\alpha'\|$. Conversely, α' is uniformly continuous; for $\|P\|$ small, $\|\alpha'(t)-\alpha'(x_i)\|<\epsilon$ on $[x_{i-1},x_i]$, so

$$\int_{x_{i-1}}^{x_i}\|\alpha'\|\le\|\alpha'(x_i)\|\Delta x_i+\epsilon\Delta x_i=\Big\|\int_{x_{i-1}}^{x_i}(\alpha'(t)+\alpha'(x_i)-\alpha'(t))dt\Big\|+\epsilon\Delta x_i\le\|\Delta\alpha_i\|+2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\|\le\Lambda(\alpha)+2\epsilon(b-a)$. ■

5.5 The Riemann–Stieltjes integral

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g,cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\le g$ implies $\int f\,d\alpha\le\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\ge0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $|\int f\,d\alpha|\le\int|f|\,d\alpha\le\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i\ge0$. For (c), $U(f,P,\alpha+\beta)=U(f,P,\alpha)+U(f,P,\beta)$ and likewise for L. ■

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f\in\mathcal{R}(\alpha)$;
(b) for every $\epsilon>0$ there is P with $U(f,P,\alpha)-L(f,P,\alpha)<\epsilon$;
(c) there is A such that for every $\epsilon>0$ some P_0 has $|S(f,P,\alpha)-A|<\epsilon$ for every $P\supset P_0$ and every choice of tags. In that case $A=\int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): with $A=\int f\,d\alpha$ and P_0 as in (b), for $P\supset P_0$ both $S(f,P,\alpha)$ and A lie in $[L(f,P,\alpha),U(f,P,\alpha)]$, an interval of length $<\epsilon$.

(c)⇒(b): for $P\supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha)\le A+\epsilon$ and $L(f,P,\alpha)\ge A-\epsilon$. ■

Example 5.3

(a) If α is the unit step at $c\in(a,b)$ ($\alpha=0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha=f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.

(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha = \sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation, write $\alpha = \alpha_1 - \alpha_2$ with α_i increasing (Theorem 5.16). If $f \in \mathcal{R}(\alpha_1) \cap \mathcal{R}(\alpha_2)$ for some such decomposition, we say $f \in \mathcal{R}(\alpha)$ and set $\int f\,d\alpha = \int f\,d\alpha_1 - \int f\,d\alpha_2$; by Theorem 5.22 below this does not depend on the decomposition. The Riemann–Stieltjes sum $S(f, P, \alpha)$ is defined as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i - m_i < \epsilon$ for all i; then $U(f, P, \alpha) - L(f, P, \alpha) \le \epsilon(\alpha(b) - \alpha(a))$. ∎

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f, P, \alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f, P, \alpha) = S(f, P, \alpha_1) - S(f, P, \alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$; we show $f \in \mathcal{R}(V)$. Let $K = \sup|f|$ and $\epsilon > 0$. Choose P_0 as in (b) and refine it so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (definition of $V(b)$; refining keeps this). For $P \supset P_0$ write

$$U(f, P, V) - L(f, P, V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

Since $\Delta V_i \ge |\Delta\alpha_i|$, the first sum is at most $2K\epsilon$. For the second, choose s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, and swap them where $\Delta\alpha_i < 0$; then the second sum is less than $\sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f, P, \alpha) - S'(f, P, \alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$, where S, S' are sums with tags t_i, s_i. By Theorem 5.20, $f \in \mathcal{R}(V)$. Then the sums $S(f, P, V - \alpha) = S(f, P, V) - S(f, P, \alpha)$ satisfy the condition of Theorem 5.20(c) for the increasing function $V - \alpha$, so $f \in \mathcal{R}(V - \alpha)$, and with the decomposition $\alpha = V - (V - \alpha)$, $\int f\,d\alpha = A$. ∎

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a, b]$. If f is bounded and $f \in \mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i = \alpha'(u_i)\Delta x_i$ with $u_i \in (x_{i-1}, x_i)$, so

$$\left| S(f, P, \alpha) - \sum f(t_i)\alpha'(t_i)\Delta x_i \right| \le K \sum |\alpha'(u_i) - \alpha'(t_i)|\Delta x_i.$$

By uniform continuity of α' this is $\le K\epsilon(b - a)$ once $\|P\|$ is small. Combining with Theorem 5.22(b), the Riemann sums of $f\alpha'$ over all sufficiently fine refinements of some P_0 are within $\epsilon + K\epsilon(b - a)$ of $\int f\,d\alpha$. By Theorem 5.20 with $\alpha(x) = x$, $f\alpha' \in \mathcal{R}$ with the stated integral. ∎

Theorem 5.24 (5.5.6, integration by parts)

Let f, α be of bounded variation on $[a, b]$. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Let $A = \int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q \supset P_0$ be a partition $a = y_0 < \cdots < y_m = b$ with tags $s_k \in [y_{k-1}, y_k]$. By summation by parts,

$$S(\alpha, Q, f) = \sum_k \alpha(s_k)(f(y_k) - f(y_{k-1})) = f(b)\alpha(b) - f(a)\alpha(a) - \sum_{k=0}^{m} f(y_k)(\alpha(s_{k+1}) - \alpha(s_k)),$$

with $s_0 = a$, $s_{m+1} = b$. The last sum is a Riemann–Stieltjes sum $S(f, P, \alpha)$ for the partition P formed by the points s_k together with the y_k (zero-length intervals contribute nothing), with tags y_k; $P \supset Q \supset P_0$. So $|S(\alpha, Q, f) - (f(b)\alpha(b) - f(a)\alpha(a) - A)| < \epsilon$, and Theorem 5.22 applies to α and f. ■

Corollary 5.25 (5.5.7, second mean value theorem)

If $f : [a, b] \to \mathbb{R}$ is increasing and g is continuous, there is $c \in [a, b]$ with $\int_a^b fg = f(a)\int_a^c g + f(b)\int_c^b g$.

Proof. Let $G(x) = \int_a^x g$, a C^1 function with $G' = g$. $G \in \mathcal{R}(f)$ (Proposition 5.21), so $f \in \mathcal{R}(G)$ and $\int f\,dG = f(b)G(b) - \int G\,df$ (Theorem 5.24), while $\int f\,dG = \int fg$ (Theorem 5.23). With m, M the minimum and maximum of G, $m(f(b) - f(a)) \le \int G\,df \le M(f(b) - f(a))$, so $\int G\,df = G(c)(f(b) - f(a))$ for some c by the intermediate value theorem. Hence $\int fg = f(b)G(b) - G(c)(f(b) - f(a)) = f(a)G(c) + f(b)(G(b) - G(c))$. ■